HEALTHCARE ANALYTICS · SYNTHETIC DATA

Data Quality Monitoring Dashboard for Claims & Encounters Data

A profiling and monitoring framework for catching nulls, anomalies, and inconsistencies in large healthcare datasets before they reach downstream reporting and HEDIS measures.

8 FIELDS MONITORED	**95%** TARGET PASS-RATE	**5.4%** WORST FIELD NULL RATE	**16wk** TREND WINDOW TRACKED

THE PROBLEM

Large claims and encounters datasets accumulate quality issues silently — a missing diagnosis code, an inconsistent eligibility span, a malformed provider ID — that don't surface until a downstream report breaks or a HEDIS measure comes back wrong. Catching these early, at the field level, is far cheaper than catching them after they've propagated into client-facing reporting.

APPROACH

- Designed a profiling layer tracking null/missing rates across 8 critical fields (Member ID, Diagnosis Code, Provider NPI, Service Date, Claim Amount, Plan Code, Eligibility Span, Place of Service)
- Flagged fields exceeding a 3% null-rate threshold for investigation
- Tracked a rolling weekly composite validation pass rate against a 95% target to catch degradation early

FINDINGS

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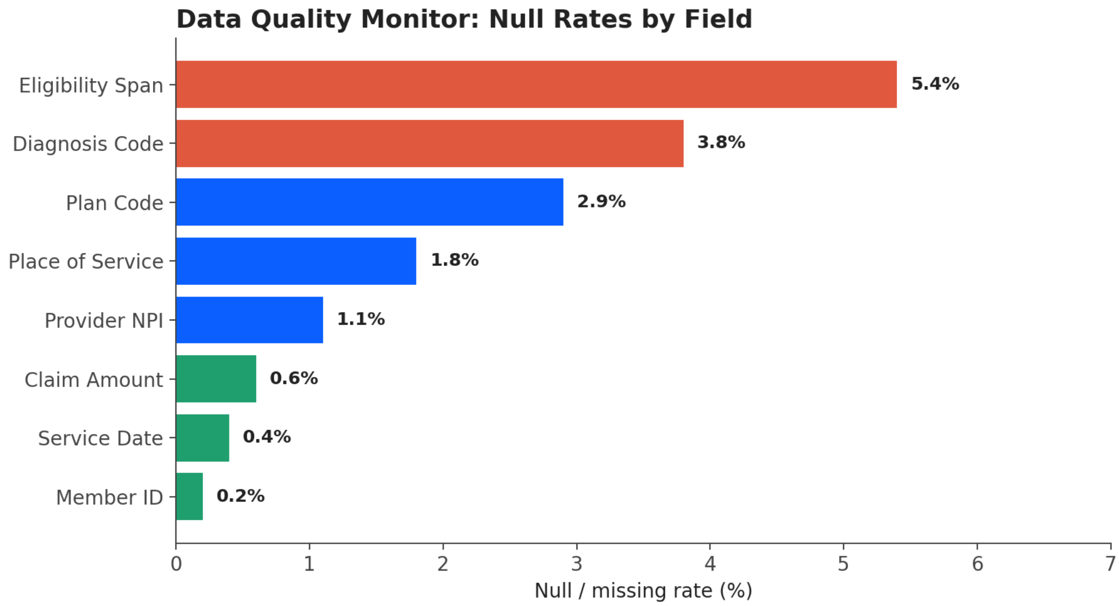
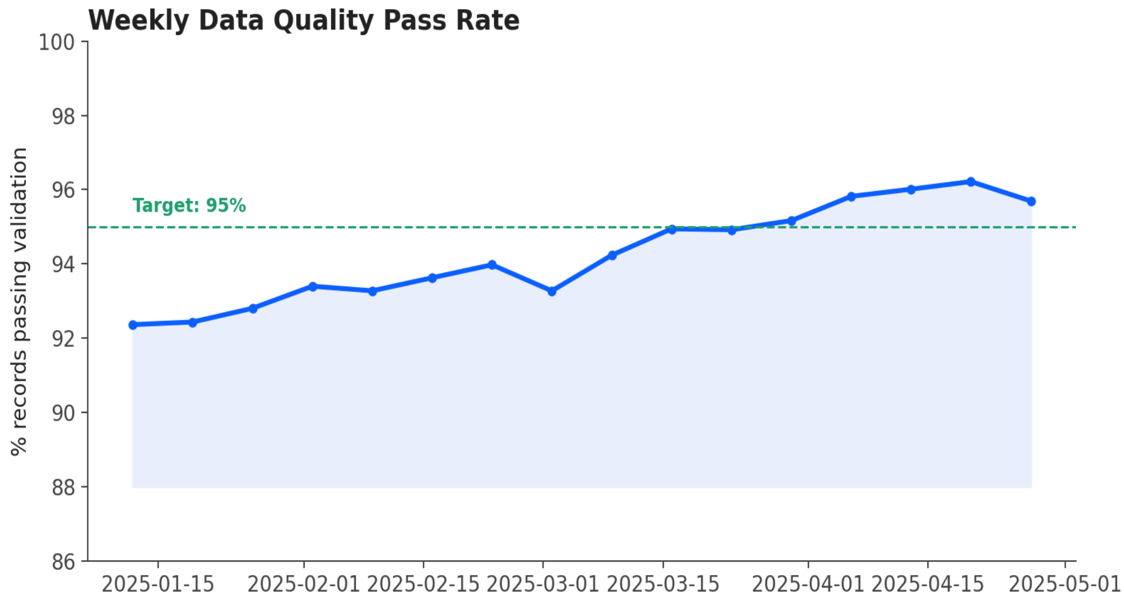


Eligibility Span and Diagnosis Code carried the highest null rates — both tied to upstream source system timing issues rather than true data loss.



The composite pass rate trended upward over the 16-week window and crossed the 95% target by week 11, following upstream fixes to eligibility span logic.

TAKEAWAY

Field-level monitoring turns “the data looks off” into a specific, trackable signal. Once Eligibility Span was identified as the main driver of failed validations, the fix was targeted at the source rather than patched downstream — the difference between a one-time fix and a recurring monthly firefight.

Note on data: This project is inspired by data quality and profiling work in healthcare claims analytics, but all figures, field names, and trends shown here are illustrative and synthetic. No real client data, table structures, or volumes are represented.

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